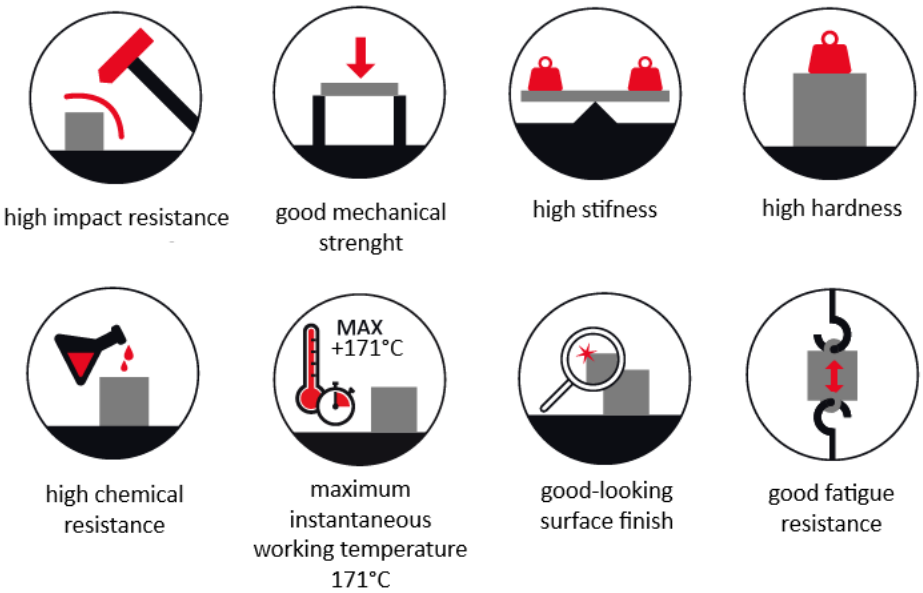
NOCTUO

PRODUCT CARD

PA12CF15



1. GENERAL INFORMATION ABOUT THE PRODUCT

Filament made of durable polyamide 12, strenghtened with carbon microfibers. Received in this way, the composite gains greater stiffness and hardness while maintaining high impact strength and layer bondability and very high temperature stability of polyamide. The perfect material for technical applications – prints of functional parts or tools. It has a very low moisture absorption for polyamides and relatively little shrinkage.

Main PA12CF15 features:

- high mechanical strength, stiffness and hardness,
- high impact resistance,
- good fatigue strength,
- high warm resistance,
- high chemical resistance, especially to organic solvent, including oils, greases and fuels.

2. TECHNICAL PARAMETERS

- Due to the high sensitivity to the parameters and orientation of the model in printing, the results for the composite itself (obtained on injected samples) are presented, which can be treated as a potential optimization target. It is possible to obtain models with a strength very close to the injection parameters. In particular, by manipulating the model and parameters, you can shape the mechanical properties of the print by manipulating the filling or fiber orientation in the model.
- The parameters given in the tables were obtained after conditioning for 40 hours at 23°C and relative humidity 50%

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ISO			
PHYSICAL				
Density	1183	-	g/cm^3	1,06
MECHANICAL				
Yield strength	527-1,-2	50mm/min	MPa	125
Elongation at break	527-1,-2	50mm/min	%	4.9
Young's modulus	527-1,-2		MPa	8900
Notched Charpy impact strength	179-1	1eA	kJ/m^2	16
Impact strength without notch according to Charpy	179-1	1eU	kJ/m^2	60
THERMAL				
Vicat softening point	306	50N	°C	171
Deflection temperature under load	75-1,-2	0,45 MPa 1,8 MPa	°C	163 159

Conditioned state - fittings conditioned at 23°C and 50% relative humidity until stability
Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

The hygroscopicity must be considered. Most problems with printing is caused by moisture filament. You may find some issues with proper adhesion too. Using perforated bed or specialized coating is recommended. Hardened nozzles should be used, due to the content of carbon fibers, which quickly wear down the nozzles from soft materials

Recommended parameters of printing:

Hotend temperature	265 - 290 °C
Bed temperature	> 100 °C
Print speed	< 200 mm/s

Drying:

Vacuum or freeze drying if possible. In conditions of drying with warm air flow, heat soaking in temperature of about 80 °C with thermocirculation. The highest drying efficiency is achieved by unwinding the required amount of filament from the spool and drying loose coils.

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
PA12CF15 needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

In proper using conditions, the product doesn't endanger health.

It's forbidden to set fire or exceed decomposition temperature!

Decomposition is typically over 350 °C.

Detailed safety information available in SDS.